

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY (CSPIT)

OCCE3003: Blockchain and its Applications (NPTEL)

Question Bank (Week 1 – Week 12)

Week 1:

1. What is Blockchain technology? Explain the basic structure of a blockchain with a diagram.
2. Differentiate between centralized, decentralized, and distributed systems.
3. Explain the key characteristics of blockchain such as transparency, immutability, and decentralization.
4. What is a cryptographic hash function? Explain how hashing ensures data integrity in blockchain.
5. Compare traditional databases vs blockchain systems in terms of architecture and trust model.

Week 2:

1. Explain the role of public key cryptography in blockchain systems.
2. What are digital signatures? Explain the signing and verification process.
3. Describe the Merkle Tree structure and its importance in blockchain.
4. Explain how hash pointers help maintain the integrity of blockchain data.
5. Differentiate between the following cryptographic concepts:
 - Hash function
 - Digital signature
 - Public key encryption

Week 3:

1. Explain the working of Bitcoin transactions with a block structure.
2. What is Proof of Work (PoW)? Explain how miners validate transactions.
3. Define the following Bitcoin terms:
 - Miner
 - Block reward
 - Nonce
 - Mining difficulty
4. Explain the double-spending problem and how blockchain solves it.
5. What is a Bitcoin wallet? Explain different types of wallets.

Week 4:

1. What is a consensus mechanism in blockchain?
2. Compare the following consensus algorithms:
 - Proof of Work (PoW)
 - Proof of Stake (PoS)
3. Explain the Byzantine Generals Problem and its relevance in distributed systems.
4. What is Practical Byzantine Fault Tolerance (PBFT)?
5. Discuss advantages and disadvantages of Proof of Work consensus.

Week 5:

1. What is a smart contract? Explain its working with an example.
2. Explain the architecture of the Ethereum blockchain.
3. What is Gas in Ethereum? Why is it required?
4. Explain the difference between:
 - Bitcoin blockchain
 - Ethereum blockchain
5. Describe the lifecycle of a smart contract deployment.

Week 6:

1. Discuss the use of blockchain in supply chain management.
2. Explain how blockchain can improve healthcare data management.
3. What is Decentralized Finance (DeFi)?
4. Explain the role of blockchain in identity management systems.
5. Discuss advantages and limitations of blockchain in real-world applications.

Week 7:

1. What is a permissioned blockchain? How does it differ from a permissionless blockchain?
2. Explain the concept of access control in permissioned blockchain networks.
3. Compare public, private, and consortium blockchains with suitable examples.
4. What are the advantages of permissioned blockchains in enterprise applications?
5. Explain the role of identity management in permissioned blockchain systems.

Week 8:

1. Describe the architecture of Hyperledger Fabric.
2. Explain the role of the following components:
 - Peer nodes
 - Ordering service
 - Membership Service Provider (MSP)
3. What is a channel in Hyperledger Fabric? Why is it important?
4. Explain the transaction lifecycle in Hyperledger Fabric.
5. What are chaincode and endorsement policies?

Week 9:

1. Compare PBFT, Raft, and Kafka-based consensus mechanisms.
2. Explain how consensus is achieved in Hyperledger Fabric.
3. What is finality in blockchain consensus? Why is it important?
4. Analyze the scalability limitations of PBFT in large networks.
5. Explain the concept of leader-based consensus algorithms.

Week 10:

1. What are the major scalability challenges in blockchain systems?
2. Explain the concept of sharding in blockchain.
3. What are Layer-2 solutions? Explain with examples.
4. Compare on-chain vs off-chain scaling techniques.
5. Explain how sidechains improve blockchain scalability.

Week 11:

1. What are the common security threats in blockchain networks?
2. Explain the 51% attack and its implications.
3. What is double spending attack? How is it prevented?
4. Explain the role of Zero-Knowledge Proofs (ZKP) in privacy.
5. What are mixing services and ring signatures?

Week 12:

1. Discuss blockchain applications in:
 - Supply chain
 - Healthcare
 - Finance
2. What is Decentralized Finance (DeFi)? Explain its components.
3. Explain the concept of Non-Fungible Tokens (NFTs).
4. What are the challenges in adopting blockchain in real-world systems?